

Business Analytics – SAS Base

Lecture 2: Import data and data manipulation

Aim of the lecture

- Introduce you to different data file formats and teach you how to import data into SAS
- Walk you through the main data manipulation functions

Agenda

- **Data file intro**
- Import data into SAS
- Data manipulation functions
- Recap

Data could be stored in different formats

File format	Extension	Main Platform
Excel	xls, xlsx	Excel, can be imported
SAS dataset	sas7bdat	SAS
Comma Separated Values file	csv	Excel?, can be imported to various platforms
Tab delimited file	txt	Can be imported to various platforms

Let us look at some examples

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PROC IMPORT is the SAS procedure that can import data files into SAS

PROC IMPORT

DATAFILE="filename" (path and name of the input file)

OUT=SAS-data-set (name chosen by you)

DBMS= identifier (XLS, XLSX, CSV, etc.) **REPLACE;**

SHEET="Sheet-name" (only for Excel files);

GETNAMES=YES;

DATAROW=N;

RANGE="range-name";

RUN;

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Data manipulation functions can revise the data based on the need of the analysis

Area	Functions
Variable selection	KEEP, DROP
Data selection	WHERE, IF
Derived logic	IF-THEN-ELSE, create new variable
Deal with missing value	MISSING, . for number, ' ' for character

This is a list to help you memorize, details will follow

DROP function deletes variables from data

Method 1

```
DATA data_output;  
SET data_input ;  
DROP variables to be dropped;  
RUN;
```

Method 2

```
DATA data_output;  
SET data_input (DROP = variables to be dropped);  
RUN;
```

KEEP function keeps selected variables from data

Method 1

```
DATA data_output;  
SET data_input ;  
KEEP variables to be kept;  
RUN;
```

Method 2

```
DATA data_output;  
SET data_input (KEEP = variables to be kept);  
RUN;
```

WHERE or IF function narrows down the data you are interested in

Method 1

DATA data_output;

SET data_input ;

WHERE the logic you would like to use for data selection;

RUN;

Method 2 (not very popular)

DATA data_output;

SET data_input;

IF the logic you would like to use for data selection;

RUN;

New variable can be directly created with the name you choose

Method

```
DATA data_output;
```

```
SET data_input ;
```

```
new_variable = your logic for the new variable;
```

```
FORMAT new_variable format you want; (Optional)
```

```
RUN;
```

IF-THEN-ELSE is useful for derived logics

Method

DATA data_output;

SET data_input ;

IF logic 1 THEN action based on logic 1;

ELSE IF logic 2 THEN action based on logic 2;

ELSE action for the leftover scenarios;

RUN;

Missing value is an important topic in **real life projects**

Method 1

```
DATA data_output;
```

```
SET data_input ;
```

```
WHERE variable = . (for numeric variable)/ " (for character variable);
```

```
RUN;
```

Method 2

```
DATA data_output;
```

```
SET data_input
```

```
WHERE MISSING(variable);
```

```
RUN;
```

Recap: Now you know how to import data file into SAS and manipulate the data

- There are different types of data files (xls, csv, sas7bat, txt, etc.)
- PROC IMPORT can import data files into SAS as a SAS dataset
- Various data functions can be leveraged to manipulate the data (variable selection, data selection, derived logic, and deal with missing variables)

*Thank
you*

